

Breathing to live



USEFUL SIDE EFFECT

The body uses the airflow from breathing in several ways. One is to blow air into a confined space, which makes the air and container vibrate. If that container is a wind instrument, like Charlie Parker's saxophone or Miles Davis's trumpet, it can produce inspiring sounds. Great breath control is needed from chest muscles ruled by the musician's brain.

GALEN'S TEACHINGS

Influential Roman physician Claudius Galen enlarged on the breathing notions of the Greeks. He devised a complex theory of breathing based on the work of several earlier writers. He thought that *pneuma*, a natural "world spirit," seeped from the air down the windpipe into the lungs, along hollow pulmonary veins and into the heart, where it added special life to the vital flame. This "vital spirit" was then distributed through the arterial system. The air picked up waste gases of combustion from the burning flame and discharged them in the lungs.

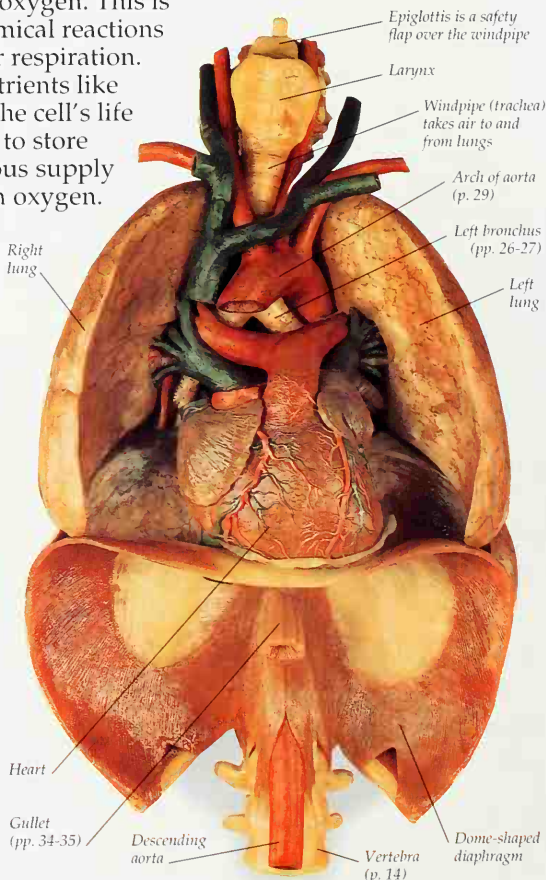


BREATHING AND BURNING

English lawyer-turned-physician John Mayow (1640-1679) carried out many studies on breathing. He proved that the muscles of the chest and diaphragm made the lungs stretch and expand like sucking bellows to draw in air. He also showed that breathing out was a nonmuscular process, because of the lungs' natural elastic recoil. By putting animals and flames in sealed glass jars, singly and together, Mayow demonstrated that they used the same fraction of air, showing a similarity between breathing and burning. The fraction was later identified and called oxygen.



WHY DOES THE LIVING BODY DIE if it cannot breathe? Breathing has a history of mystery. Ancient Greeks such as Plato and Aristotle thought that food was burned in the heart as a "vital flame," which created life and warmth. Breathing cooled the flame, and prevented it from burning too high and consuming the body. There was not much progress in the study of breathing until William Harvey described the workings of the heart (pp. 28-29). Later, tiny blood vessels and air sacs were discovered in the lungs. Scientists such as Robert Boyle, Robert Hooke, and John Mayow investigated breathing using chemistry and the mechanical principles of the bellows and suction pump. It is now known that the lungs work like a bellows to absorb oxygen. This is needed for a set of bodily chemical reactions in every cell known as cellular respiration. This process "slow-burns" nutrients like sugars, to liberate energy for the cell's life processes. The body is unable to store oxygen, so it needs a continuous supply from the air, which is one-fifth oxygen.



THE COMPONENTS OF THE CHEST

The chest is the upper of the torso's two major internal compartments; the lower is the abdomen. The dome-shaped diaphragm divides the two. Inside its flexible cage of vertebrae, ribs, and breastbone are those most vital of organs, the heart and lungs. The lungs absorb oxygen from breathed-in air, which diffuses into the blood, and the heart pumps this blood into the aorta to distribute oxygen throughout the body.